RAID 1：用镜像保留第二份数据

中文理解：副本为什么有用

RAID 1 把同一逻辑块保存到两个物理位置。单盘故障时，存活副本仍能提供原来的内容；随后可用它重建替换盘。这个结论依赖副本已正确同步，以及存活盘能够读出。

容量与手算例

沿用课件记号：N 是原始数据所需的等容量盘数，C 是每盘容量。RAID 1 用 2N 块盘提供约 N×C 可用容量。自编例：N=2、C=4 TB，需要 4 块物理盘，可用容量约 8 TB，忽略格式等开销。

易错点与边界

能恢复任意单盘失效，不代表能恢复任意两盘失效。两盘若属于同一个镜像对，就可能丢数据。镜像也不是备份：误删除、错误写入或共同故障仍需要其他保护。

考试英文表述

RAID 1 stores a mirrored copy of each logical block. If one disk fails, a surviving synchronized mirror can supply the data and rebuild a replacement. With N disks' worth of data, two-way mirroring requires 2N equal-capacity physical disks, so usable capacity is about half of raw capacity. Mirroring protects against disk failure; it does not replace an independent backup.

Q1. Why can RAID 1 survive a single-disk failure?

Each logical block has a copy on another disk. Assuming the mirror was synchronized and remains readable, the system can use that copy after one disk fails.

Q2. What is the main capacity tradeoff, and is mirroring a backup?

Two-way mirroring uses about twice the physical capacity needed for one copy. It is not a substitute for backup, because accidental deletion or corrupted writes may affect both copies.

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RAID 4：用 XOR 校验代替完整复制

中文理解：按条带计算校验

同一条带中，P = D1 XOR D2 XOR ... XOR DN。这里 XOR 是按位异或，不是普通加法。某个数据块丢失时，用 P 与其余块异或，就能消去已知贡献；若校验盘坏了，也能从数据重算 P。

恢复例与适用条件

自编例：A=10、B=11、P=01，丢失 B 后以 A XOR P 恢复为 11。假设其余块正确且可读、故障位置已知、校验与数据一致。N 块数据盘只需额外一块校验盘；每条带一个关系一般不足以解两个未知块。

容量、性能与易错点

N+1 块等容量物理盘提供约 N 块盘的可用空间。条带化可让数据访问并行；但所有正常数据写入都涉及专用校验盘更新，因此并行写受它限制。校验不是额外保存一份完整文件，也不保证任何读写都更快。

考试英文表述

RAID 4 stripes data across N disks and stores one XOR parity block per stripe on a dedicated parity disk. A single missing block can be reconstructed from parity and the surviving blocks, assuming they are correct and consistent. It requires N+1 equal-capacity disks for N disks' worth of data. Striping can improve parallel access, but parity updates concentrate write traffic on one disk.

Q1. How does XOR parity recover a missing data block?

XOR the parity block with every surviving data block in that stripe. Equal contributions cancel, leaving the missing block. This assumes one missing block and correct, consistent surviving information.

Q2. Why can the dedicated parity disk become a bottleneck?

Writes to different data disks still require updates to the same parity disk. That shared update path limits concurrent write throughput even when the data disks could work in parallel.

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RAID 5：把校验块的负担分散到各盘

中文理解：改位置，不改校验关系

RAID 5 仍然为每条带保存一份 XOR 校验，但校验块随条带分布到不同磁盘。图中的灰块轮换位置，所以不存在一块专门承接所有校验更新的盘。分散校验缓解的是 RAID 4 的集中写入负载。

容量、性能与例子

N+1 块等容量盘提供约 N 块盘的可用容量。自编例：4 块 4 TB 盘，N=3，可用容量约 12 TB。数据访问仍可利用条带并行；校验更新开销仍存在，性能提升取决于负载，并非任何一次写入都更快。

易错点与边界

校验分布在多块盘上，不代表每条带有多条独立校验方程；RAID 5 仍然只保证单盘失效恢复。若同一条带有两个未知数据块，一份 XOR 校验通常不足以唯一解出它们。

考试英文表述

RAID 5 distributes parity blocks across the disks instead of using a dedicated parity disk. This spreads parity-update traffic across different stripes and avoids RAID 4's single parity-disk hotspot. It still stores one parity block per stripe, requires N+1 disks for N disks' worth of data, and tolerates one disk failure. Distributed parity does not remove update overhead or provide two-disk fault tolerance.

Q1. What does RAID 5 improve over RAID 4?

It distributes parity blocks for different stripes across disks, spreading parity-update traffic. This avoids concentrating all parity writes on one dedicated disk, although parity still has to be maintained.

Q2. Does distributing parity allow recovery from two disk failures?

No. Each stripe still has only one independent XOR parity relation. Two missing blocks in a stripe generally cannot be uniquely reconstructed from that single relation.

方案	物理盘数 / 可用容量	主要取舍
RAID 1	2N / N×C	完整镜像；容量开销大
RAID 4	N+1 / N×C	专用校验盘；写入集中
RAID 5	N+1 / N×C	分布校验；仍需更新校验

N：原始数据所需等容量盘数；C：单盘容量。三者均可恢复任意单盘失效；均不代替独立备份。

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